



Big Data Mining Final Report

An Empirical Analysis of Recommendation Systems on the Million Song Dataset



電資四 111820006 陳羿錦
資訊四 111590009 陳世昂

The Objective: A Head-to-Head Comparison of Recommendation Algorithms

To build a music recommendation system and empirically evaluate the performance of three principal algorithmic families on a large, real-world dataset.

The Contenders



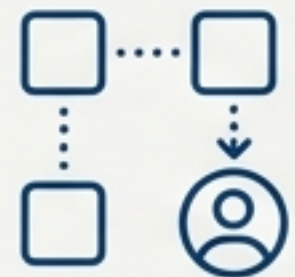
Content-Based Filtering:

Recommends items based on their intrinsic properties (audio and text features).



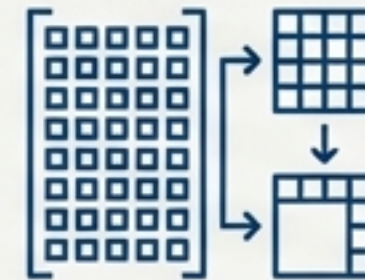
User-User Collaborative Filtering:

Finds similar users to make recommendations.



Item-Item Collaborative Filtering:

Recommends items similar to those a user has liked.



Matrix Factorization:

Uncovers latent features of users and songs.



Hybrid Model:

A weighted ensemble of the four core models.

The Challenge: The Million Song Dataset is Immense and Incredibly Sparse



The Core Problem: A staggering sparsity of **99.9878%**. This means the user-song interaction matrix is almost entirely empty, posing a monumental challenge for algorithms that rely on finding patterns in user behavior.

A 'Long Tail' of Interaction Dominates the Dataset

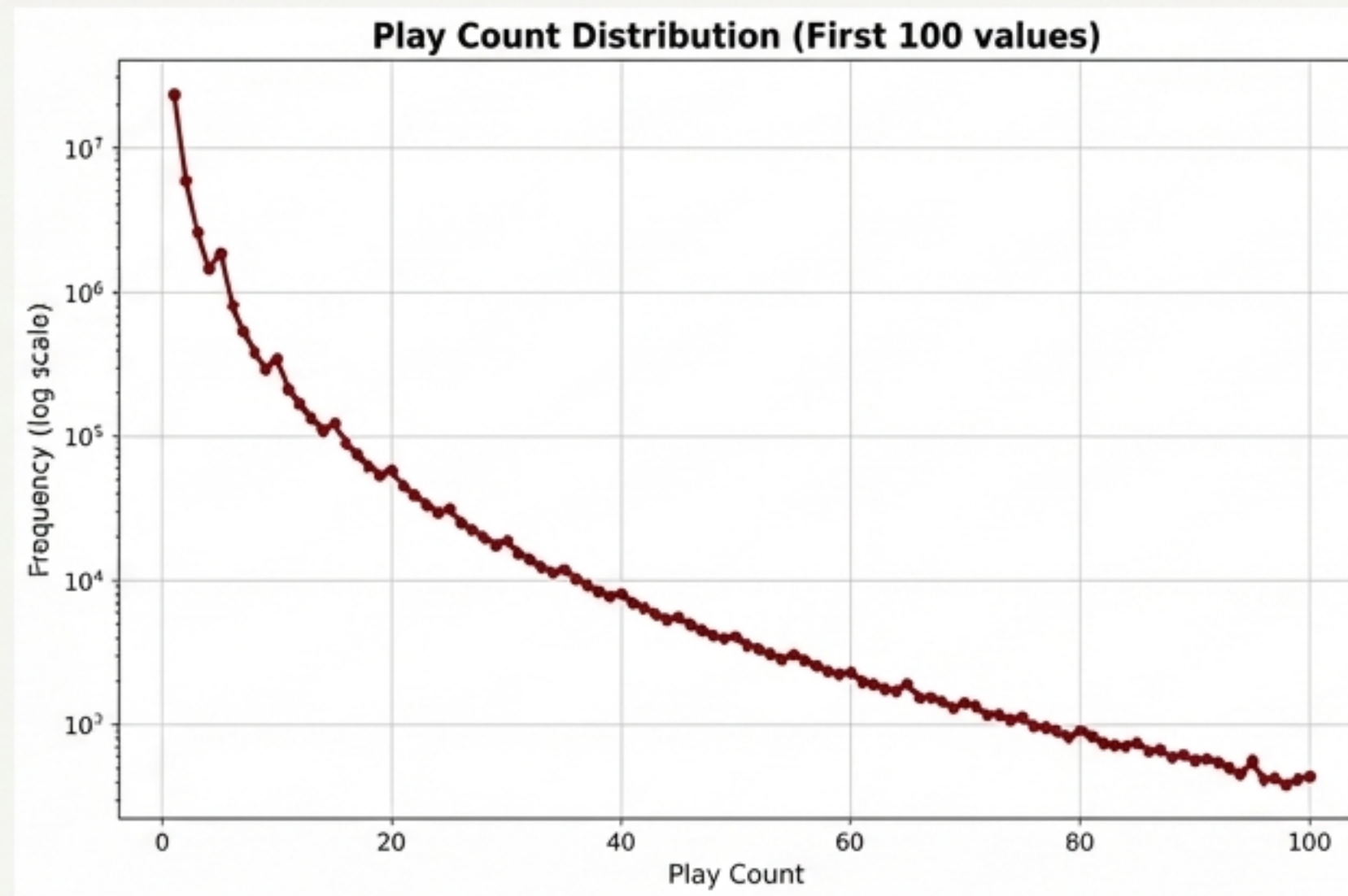
Key Finding

The vast majority of listening activity is fleeting. A detailed analysis of play counts reveals:

- **59.42%** of all interactions are from songs played only **once** by a user.
- Over **89%** of interactions come from songs played five times or fewer.

Implication

Most user-song pairings lack the repeated engagement needed for many algorithms to learn strong preferences.



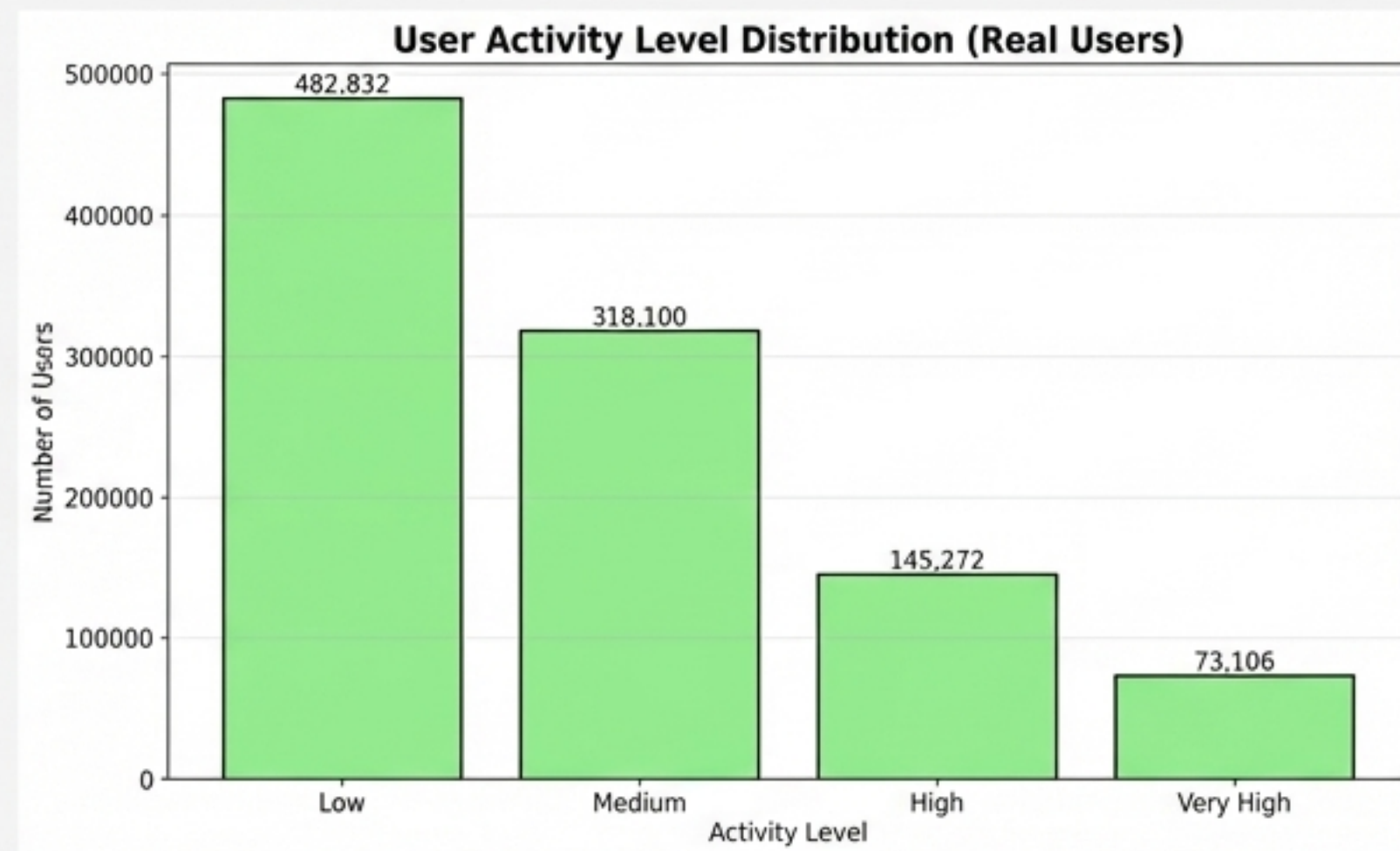
The User Base is Dominated by Low-Activity Listeners

Key Insight

The distribution of user activity is heavily skewed. The largest segment of the user population engages with a very small number of songs.

Implication

Finding 'similar users' is extremely difficult when most users have very little listening history to compare.



User Segmentation by Activity:

- **Low (1-20 songs):** 482,832 users (**Largest group**)
- **Medium (21-50 songs):** 318,100 users
- **High (51-100 songs):** 145,272 users
- **Very High (101+ songs):** 73,106 users

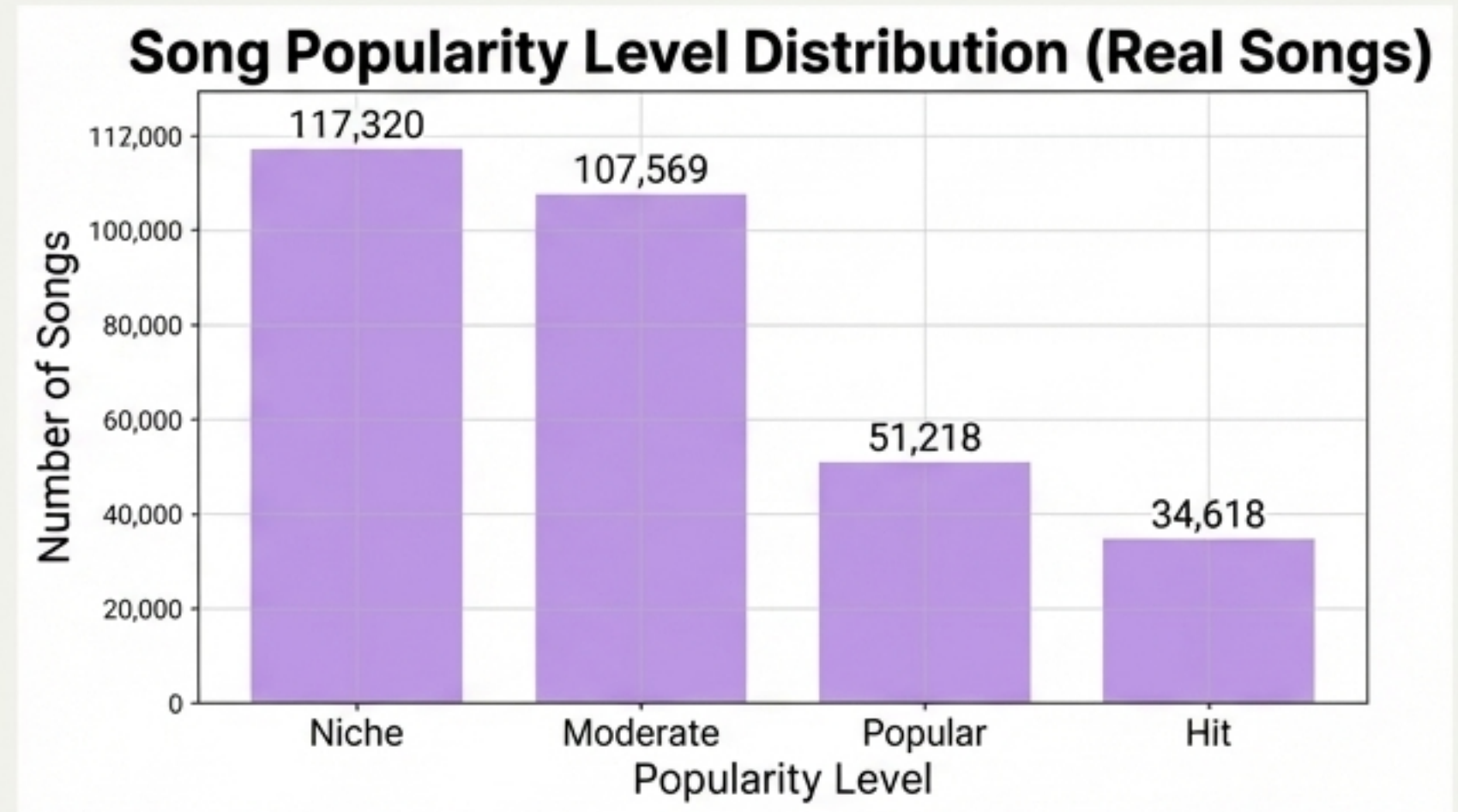
The Music Catalog is Primarily Composed of Niche and Obsiure Tracks

Key Insight

Mirroring the user activity, the song catalog is not filled with mainstream hits. The majority of songs have very few listeners.

Implication

With most songs having a small audience, it is difficult for item-based collaborative filtering to find strong co-occurrence patterns.

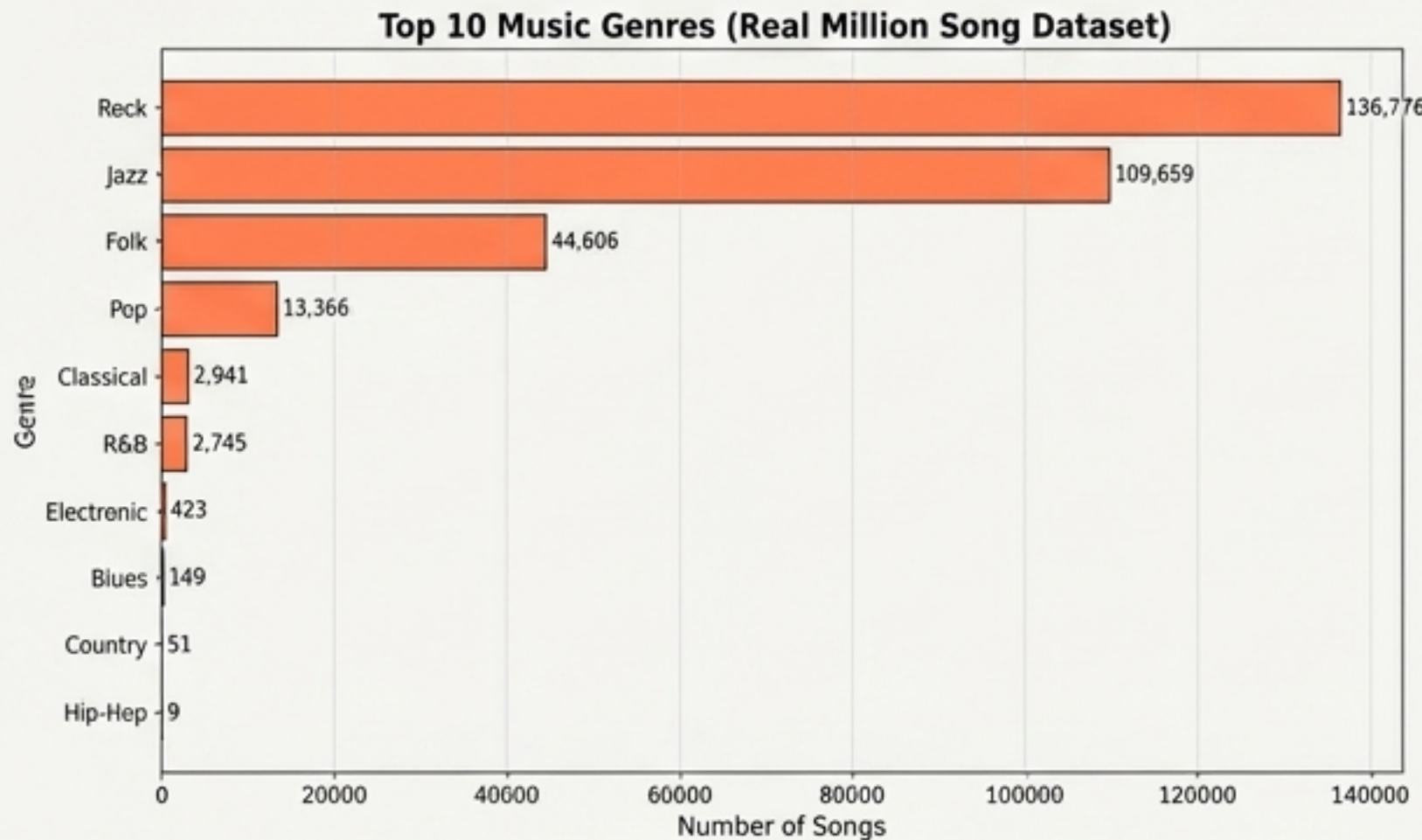


Song Segmentation by Popularity:

- **Niche:** 117,320 songs (**Largest group**)
- **Moderate:** 107,569 songs
- **Popular:** 51,218 songs
- **Hit:** 34,618 songs (**Smallest group**)

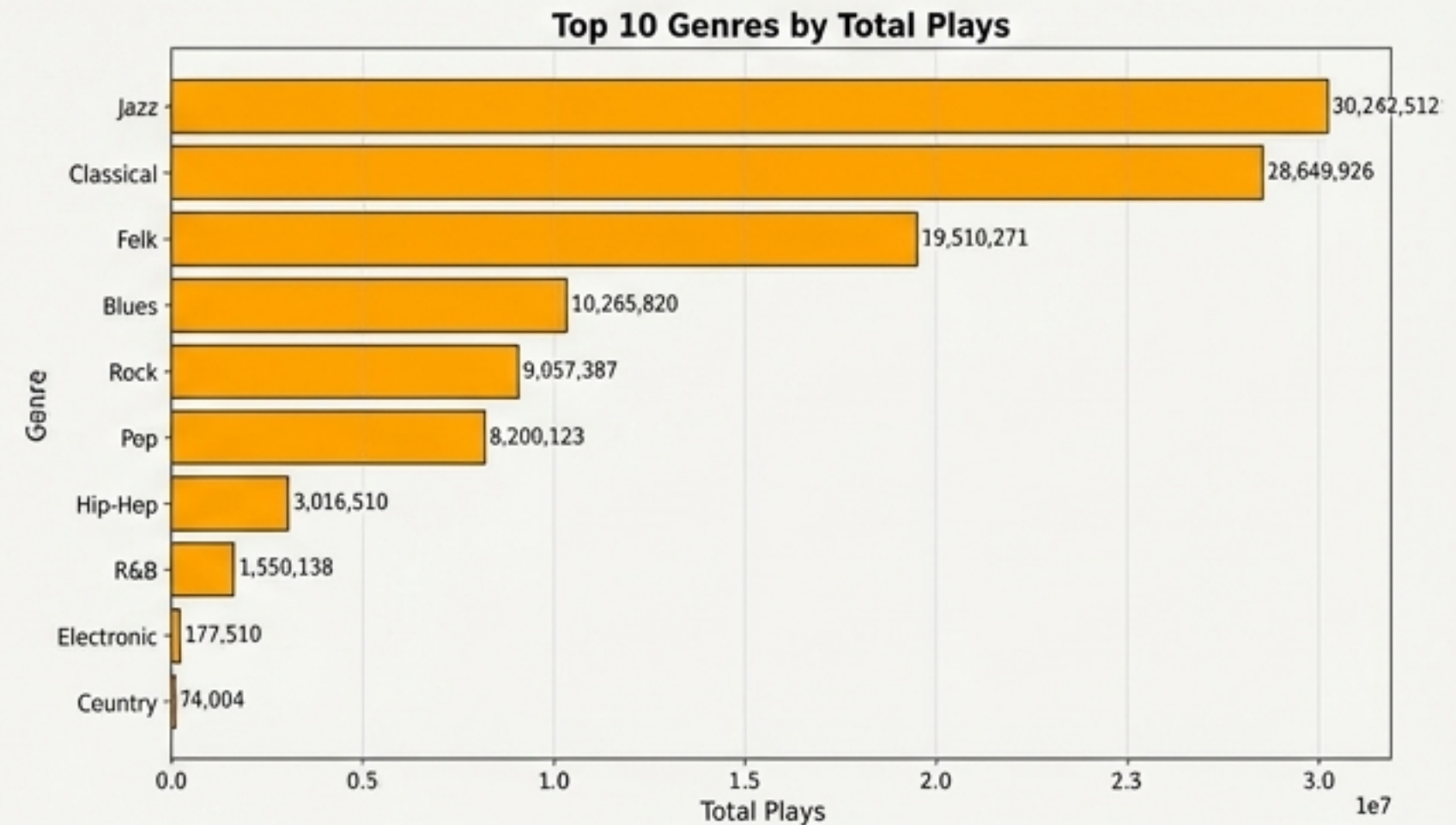
The Genre Paradox: Catalog Size Does Not Equal Listener Engagement

Top Genres by Number of Songs



****Observation**:** The catalog is primarily a rock and jazz library.

Top Genres by Total Plays



****Observation**:** Rock falls to fifth place with only 9.1 million plays.

****Key Takeaway**:** The most numerous genres in the dataset are not the most listened to. User preference is highly concentrated in specific genres, regardless of catalog size.

The Experiment: A Rigorous "Leave-One-Out" Evaluation

Evaluation Strategy

We employed a Leave-One-Out cross-validation strategy, which is better suited for extremely sparse data than a traditional train/test split.

For each user in the test set:

1. One random song is hidden from their listening history (the "target").
2. The model generates recommendations based on their remaining history.
3. Success is measured by whether the hidden "target" song appears in the top recommendations.

- **Scale:** The evaluation was performed on a sample of **5,000 users**.
- **Primary Metric: Hit Rate@10** - The percentage of users for whom the hidden song was correctly recommended within the top 10 results.



Hide 1 song ("target")



Generate Top 10 Recommendations



Measure Success: Is 'target' in the list?

A Reality Check: Sparsity's Decisive Impact on Performance

Performance of five models tested on 5,000 users using Leave-One-Out evaluation.

Model	Successful Users	Hit Rate@5	Hit Rate@10	Hit Rate@20	Coverage
Hybrid	5,000	0.98%	1.40%	1.96%	16.10%
Item-CF	5,000	0.52%	1.16%	2.22%	24.27%
Content-Based	5,000	0.16%	0.40%	0.70%	16.23%
Matrix Factorization	4,993	0.00%	0.00%	0.00%	10.87%
User-CF	4,945	0.00%	0.00%	0.00%	11.99%

Summary: In this high-sparsity environment, all models achieved very low accuracy. The Hybrid model performed best, while User-CF and Matrix Factorization completely failed to produce relevant recommendations.

Analyzing the Results: Item-CF and Hybrid Lead the Pack

Hybrid Recommender

Winner

Hit Rate@10: **1.40%**

Analysis: Achieved the highest accuracy by combining the outputs of all four models (e.g., Content-Based: 30% weight, User-CF: 25%, etc.). This ensemble approach effectively balanced the strengths of each method, providing a small but consistent performance lift.



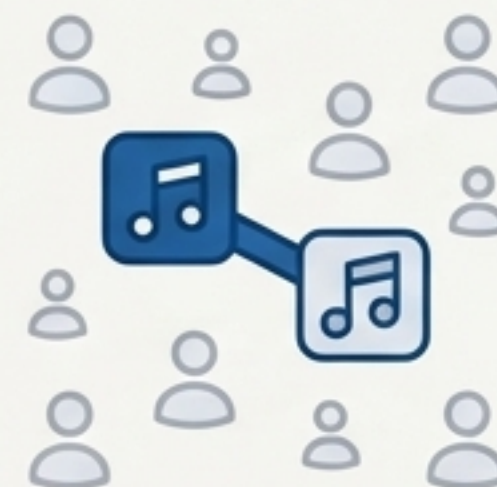
Item-Item Collaborative Filtering

Best Single Model

Hit Rate@10: **1.16%**

Coverage: 24.27% (Highest of all models)

Analysis: Its relative success stems from relying on item co-occurrence patterns, which are more robust than user-similarity patterns in a dataset dominated by low-activity users.



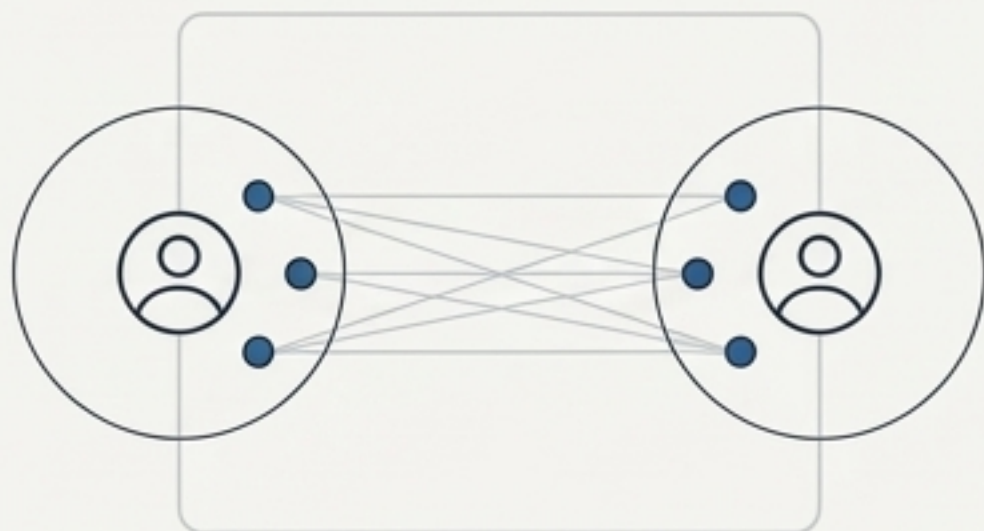
Deconstructing Failure: Why User-CF and Matrix Factorization Faltered

The Common Cause: Extreme data sparsity and the long-tail user distribution.

User-User Collaborative Filtering

Hit Rate@10: 0.00%

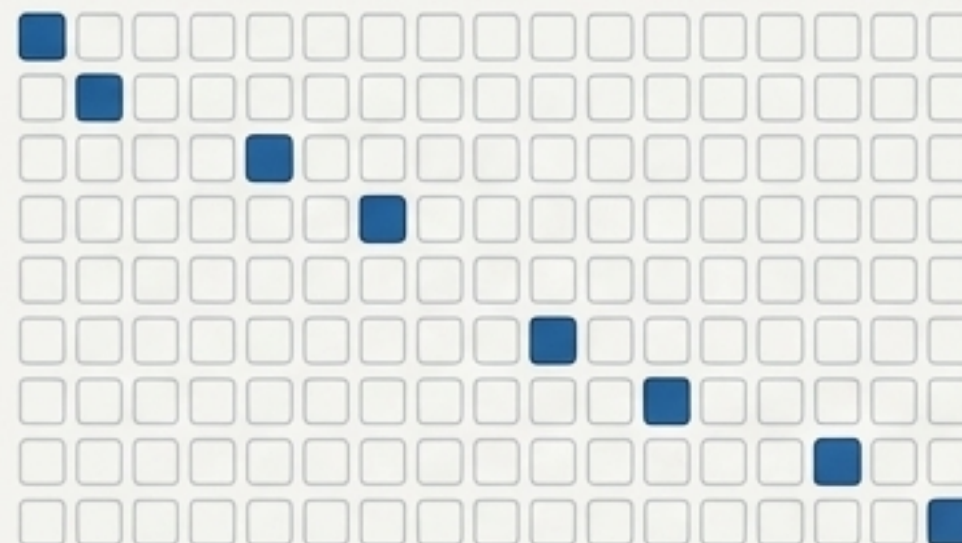
Reasoning: This model depends on finding users with overlapping tastes. With 482,832 users having listened to 20 or fewer songs, the chance of finding meaningful 'neighbors' with similar listening patterns is statistically negligible. The model simply lacks sufficient data to function.



Matrix Factorization

Hit Rate@10: 0.00%

Reasoning: While MF models are powerful, they learn latent factors from the patterns of user-item interactions. When the interaction matrix is 99.9878% empty, the model is essentially trying to learn from noise and has no significant patterns to converge upon, leading to a complete failure in the predictive task.

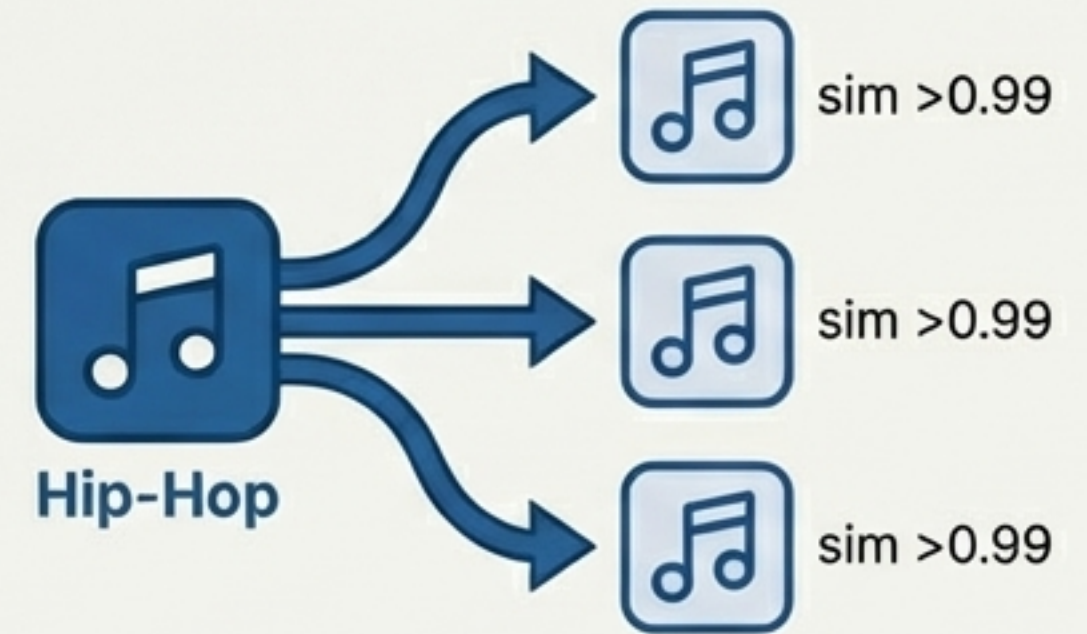


Beyond the Metrics: Qualitative Examples Reveal Underlying Potential

Key Insight: Despite low quantitative scores, individual test cases show the models were functioning as designed.

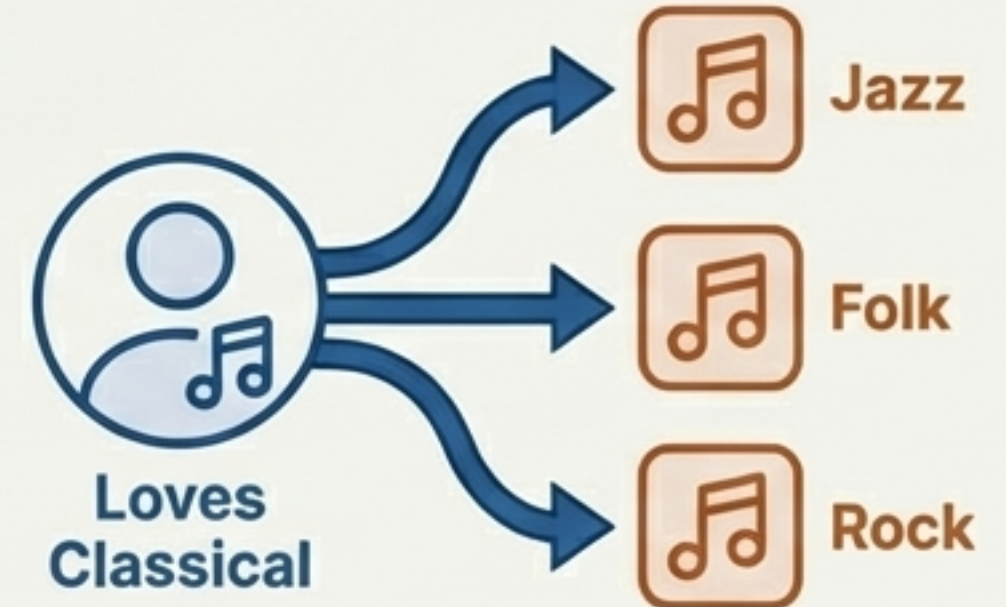
Content-Based Success Story

- **Input:** The highly popular Hip-Hop song `SOBONKR12A58A7A7E0`.
- **Output:** Successfully recommended other popular Hip-Hop tracks with similarity scores >0.99 , like `Song\_81C206F1` and `Song\_F72A7F54`.
- **Verdict:** Demonstrates a strong capability for 'find more like this' tasks, ideal for cold-start scenarios.



Matrix Factorization's Hidden Talent

- **Input:** A user profile with a strong preference for Classical music.
- **Output:** Recommended a diverse mix of **Jazz, Folk, and Rock**.
- **Verdict:** This shows the model was learning latent, cross-genre connections, showcasing its potential for personalized discovery, even though it failed the narrow leave-one-out test.



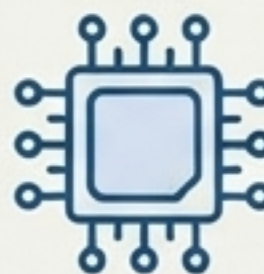
A Note on Implementation: Code Quality Was Not a Factor

Independent Code Review Summary



4.6 / 5.0
overall score

Key Strengths Highlighted



Memory Efficiency: Utilized sparse matrices (CSR format), reducing memory needs from a theoretical ~3TB to ~600MB.



Scalability: Architecture designed to handle 100M+ interactions through chunked reading, batch processing, and on-demand computations.



Innovative Evaluation: The 'Leave-One-Out' strategy was cited as a 'publication-worthy innovation' for handling extreme sparsity.

Conclusion: The observed model performance is a genuine reflection of the dataset's inherent challenges, not implementation error.

Conclusions & Key Learnings

Primary Finding

In the context of the Million Song Dataset, the **extreme data sparsity (99.9878%)** is the single most dominant factor influencing recommender performance, superseding the theoretical advantages of any single algorithm.

Key Takeaways



1. **Sparsity Favors Robustness:** In sparse environments, Item-CF is significantly more robust and effective than User-CF.



3. **Metrics are Incomplete:** Quantitative failure (like MF's 0% Hit Rate) can mask qualitative potential for personalization and discovery. A dual analysis is critical.



2. **Ensembles Provide an Edge:** Hybrid models offer a marginal but consistent improvement by hedging against the weaknesses of individual models.



4. **Data Quality over Algorithm Choice:** The results underscore that the ultimate success of a recommendation system is fundamentally constrained by the nature and density of the available data.